

LabLens

An AI-Powered AR Virtual Chemistry Lab

1. Overview

LabLens is a browser-based Augmented Reality (AR) application that allows users to perform virtual chemistry experiments using their webcam. It uses computer vision to track hand movements, enabling users to pinch, hover, and wave to interact with virtual chemicals, test tubes, and a beaker in real-time. It is designed to make learning chemistry safe, interactive, and visually engaging.

2. Architecture

The application is built using a modern, decoupled web architecture:

- Backend: Built in Python using FastAPI. It exposes REST API endpoints (/api/chemicals, /api/react, /api/examine).
- Database: A local SQLite database (lablens.db) containing over 400 chemical reactions, 118 elements, and metadata like reaction conditions, observations, and exam relevance tags (e.g., JEE).
- Frontend: Written in pure HTML, CSS, and vanilla JavaScript. It uses Google's MediaPipe Hands library directly in the browser via CDN for ultra-fast, zero-latency hand tracking.
- AI Integration: The backend integrates with Anthropic's Claude API to generate dynamic, Socratic-style examination questions based on the exact chemistry of the reaction the user just performed.

3. Modes of Operation

LabLens operates in two primary modes:

- Virtual Lab Mode: The main environment where users mix chemicals. A virtual beaker sits at the bottom of the screen. Users can open a chemical picker, load a reagent onto their hand, and drop it into the beaker to trigger reactions.
- Periodic Table Mode: An interactive, full-screen periodic table layout. Users can hover their finger over any element to view its atomic number, electronic configuration, and block classification.

4. Hand Gestures & Controls

LabLens Technical Documentation

The application utilizes precise skeletal hand tracking to recognize specific gestures:

- Pinch (Select): Touching the thumb and index finger together. Used to click the Reagents icon, switch tabs, and select a chemical.
- Hover (Drop): Moving the index finger over the 'DROP HERE' zone above the beaker. Automatically drops the loaded chemical into the beaker.
- Wave / Wash Off (Unload): Extending all five fingers and waving horizontally. Discards the currently loaded chemical from the hand.
- Fist (Reset): Curling all four fingers tightly into the palm and holding for 0.5 seconds. Instantly empties and cleans the beaker.
- 3-Fingers (Mode Toggle): Extending the index, middle, and ring fingers while holding the thumb and pinky curled for 0.5 seconds. Toggles between Lab Mode and Periodic Table mode.

5. Technical Details & Math

The UI coordinate system is meticulously designed to handle any screen size. The webcam feed is rendered natively without CSS scaling distortion (using a responsive aspect-ratio container). MediaPipe's normalized coordinates (0.0 to 1.0) are mathematically mapped to the screen's Absolute DOM boundaries, allowing for pixel-perfect intersection testing (BoundingClientRect) using the tip of the index finger as the pointer.

6. AI Examiner

When a successful reaction occurs, the user can press the 'a' key on their keyboard. The FastAPI backend sends the exact reactants, products, and observations of the reaction to Anthropic's Claude AI. The AI generates a single, highly specific Socratic question to test the user's understanding of the chemistry that just happened on their screen.